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EXPERIMENT 4: Jenkins

This experiment maps to LO1 and LO3.

Aim: To understand Continuous Integration and Jenkins installation.

Theory:

Continuous Integration (CI) is a software development practice in which developers frequently integrate their code into a shared repository. Every code change is automatically built and tested, allowing errors to be detected early in the development lifecycle. CI helps improve software quality, reduces integration problems, and speeds up software development by ensuring that new changes do not break the existing application.

The main objectives of Continuous Integration are:

- Detect bugs early in development.
- Improve collaboration among team members.
- Automate the build and testing process.
- Reduce manual errors.
- Deliver reliable software quickly.

Jenkins is an open-source automation server used to implement Continuous Integration (CI) and Continuous Delivery (CD). It automates repetitive tasks such as building, testing, and deploying applications. Jenkins integrates with various version control systems like Git and supports hundreds of plugins, making it one of the most popular DevOps tools.

Features of Jenkins

- Open-source and free to use.
- Cross-platform support (Windows, Linux, macOS).
- Large plugin ecosystem.

- Automates build, test, and deployment processes.
- Supports Pipeline as Code.
- Easy integration with Git, Docker, Kubernetes, Maven, Gradle, etc.
- Provides a web-based dashboard for monitoring jobs.

Jenkins Architecture

Jenkins mainly consists of the following components:

1. Jenkins Controller (Master)

The Controller manages the entire Jenkins environment. It schedules jobs, stores configurations, manages plugins, and communicates with build agents.

2. Jenkins Agent

Agents execute build, testing, and deployment tasks assigned by the Controller. Multiple agents can work simultaneously, allowing parallel execution of jobs.

3. Jenkins Dashboard

The dashboard is a web-based interface used to create jobs, configure pipelines, monitor builds, install plugins, and manage Jenkins.

A **Jenkins Pipeline** is a collection of automated steps that define the complete software delivery process. It automates tasks such as building, testing, and deploying applications in a Continuous Integration/Continuous Delivery (CI/CD) workflow.

Benefits of Jenkins Pipelines

- Automates repetitive tasks.
- Improves software quality.
- Provides faster feedback.
- Makes deployments consistent.
- Reduces manual effort.
- Supports version-controlled build scripts.

Types of Jenkins Pipelines

1. Declarative Pipeline

Declarative Pipeline is the recommended and easier syntax provided by Jenkins. It uses the pipeline {} block and organizes the workflow into stages and steps.

Advantages:

- Easy to read and write.
- Beginner friendly.
- Better syntax validation.
- Improved error handling.
- Structured format.

2. Scripted Pipeline

Scripted Pipeline is written using Groovy scripting language and uses the node {} block. It provides greater flexibility for implementing complex workflows.

Advantages:

- Highly flexible.
- Supports advanced scripting.
- Suitable for complex CI/CD workflows.
- Full control over pipeline execution.

Implementation:

```
C:\Users\15L>java --version
openjdk 25.0.4 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-25.0.4+7 (build 25.0.4+7-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4+7 (build 25.0.4+7-LTS, mixed mode, sharing)

C:\Users\15L>cd Downloads

C:\Users\15L\Downloads>dir
 Volume in drive C is Windows
 Volume Serial Number is 448A-211F

 Directory of C:\Users\15L\Downloads

08/07/2026  12:39 PM    <DIR>          .
08/06/2026  02:13 PM    <DIR>          ..
04/16/2026  10:47 AM         1,467,652 345_Creating_the_next_Digital_.pdf
08/06/2026  02:08 PM           829,077 Aaditya_Surve_2401119_ASDD_Experiment05.odt
07/29/2026  02:39 PM         2,008,051 AGD_EXP3.docx
01/16/2026  08:16 AM     1,199,641,816 Anaconda3-2025.12-1-Windows-x86_64.exe
08/03/2026  08:13 AM        14,492,624 apache-tomcat-9.0.120.exe
07/22/2026  02:59 PM         1,353,855 asd&d exp 2(84).docx
07/21/2026  11:28 AM           631,635 asd&d exp 2.pdf
08/04/2026  10:19 AM           211,949 ASD&D EXP-4_9767.docx
07/31/2026  03:38 PM         1,462,848 ChatGPT Installer.exe
04/15/2026  11:53 AM               159 citation-380401936.ris
08/03/2026  09:11 AM               916 Documents - Shortcut.lnk
08/03/2026  09:04 AM       582,839,767 eclipse-jee-2026-06-R-win32-x86_64.zip
07/29/2026  02:19 PM           219,916 Exp3_Krish_Kanal_2401040_T12 (1) (1).docx
07/14/2026  11:40 AM           224,179 Experiment 1_43.pdf
07/29/2026  02:32 PM           269,992 experiment3_t21_093_aryaraut.docx
07/28/2026  11:17 AM           913,917 Experiment_3_ASD&D_2401031_Atharva Gokhale (1).docx
07/28/2026  11:17 AM           913,917 Experiment_3_ASD&D_2401031_Atharva Gokhale (2).docx
07/28/2026  11:15 AM           913,917 Experiment_3_ASD&D_2401031_Atharva Gokhale.docx
07/28/2026  11:06 AM           734,916 Experiment_3_ASD&D_2401031_Atharva Gokhale.pdf
07/28/2026  11:40 AM           813,763 Experiment_3_ASD&D_2401046.pdf
07/15/2026  03:08 PM           186,197 Git_Commands_Reference.docx
08/06/2026  01:11 PM         1,500,190 J-2 (1).pdf
08/07/2026  12:22 PM         1,500,190 J-2 (2).pdf
08/03/2026  03:15 PM         1,500,190 J-2.pdf
08/04/2026  10:15 AM         1,500,190 J-2_9323.pdf08/07/2026  12:39 PM       101,103,716 jenkins.war
07/31/2026  03:38 PM             5,120 Microsoft.Services.Store.winmd
02/06/2025  10:11 AM         1,451,640 MSTeamsSetup.exe
09/19/2025  08:25 AM        31,375,360 node-v22.19.0-x64.msi
08/03/2026  03:44 PM       115,990,528 OpenJDK25U-jdk_x64_windows_hotspot_25.0.4_7.msi
04/16/2026  10:52 AM    <DIR>          ScienceDirect_articles_16Apr2026_05-22-04.451
04/16/2026  10:52 AM        36,331,366 ScienceDirect_articles_16Apr2026_05-22-04.451.zip
04/15/2026  02:10 PM         5,128,727 SEPM ASSIGNEMNT 1(t21).pdf
04/15/2026  03:09 PM         4,899,439 SEPM ASSIGNMENT 2.pdf
```

```
04/15/2026  11:34 AM            15,314 SEProblemStatement.docx
04/15/2026  11:47 AM           210,761 SEProblemStatement.pdf
04/15/2026  02:09 PM            31,569 SRS_OnlineFoodDelivery_CaseStudy.docx
04/15/2026  11:45 AM            21,157 srs_template-ieee.docx
04/15/2026  11:39 AM           385,947 srs_template-ieee.pdf
04/15/2026  01:59 PM            23,092 ssrs food delivery.odt
07/25/2025  10:19 AM       133,223,520 weka-3-8-6-azul-zulu-windows.exe
              40 File(s)  2,246,329,079 bytes
               3 Dir(s)  767,068,807,168 bytes free
C:\Users\15L\Downloads>java -jar jenkins.war --enable-future-java
Running from: C:\Users\15L\Downloads\jenkins.war
webroot: C:\Users\15L\.jenkins\war
2026-08-07 07:11:44.131+0000 [id=3]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-07 07:11:44.900+0000 [id=3]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-07 07:11:44.930+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01T01:09:18.991Z; git: c9cdc9aaa434a3665b8a53b4d1cc3684992da649; jvm 25.0.4+7-LTS
2026-08-07 07:11:45.246+0000 [id=3]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-08-07 07:11:45.272+0000 [id=3]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider (file:/C:/Users/15L/.jenkins/war/WEB-INF/lib/xstream-1.4.21.jar)
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
2026-08-07 07:11:45.508+0000 [id=3]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\15L\.jenkins found at: $user.home/.jenkins
2026-08-07 07:11:45.560+0000 [id=3]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@2063c53e{Jenkins v2.568.2,/,b=file:///C:/Users/15L/.jenkins/war/,a=AVAIL
ABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@2a9bc08f{STARTED}}
2026-08-07 07:11:45.586+0000 [id=3]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@46d9aec8{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-08-07 07:11:45.586+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@14379273{STARTING}[12.1.8,sto=0] @658ms
2026-08-07 07:11:45.587+0000 [id=67]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-08-07 07:11:45.642+0000 [id=60]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.568.2
2026-08-07 07:11:45.690+0000 [id=73]    INFO    hudson.PluginManager#loadDetachedPlugins: Upgrading Jenkins. The last running version was 2.568.1. This Jenkins is version 2.568.2.
2026-08-07 07:11:45.690+0000 [id=74]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-07 07:11:45.714+0000 [id=73]    INFO    hudson.PluginManager#loadDetachedPlugins: Upgraded Jenkins from version 2.568.1 to version 2.568.2. Loaded detached plugins (and dependencies): []
2026-08-07 07:11:45.801+0000 [id=99]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-07 07:11:47.299+0000 [id=80]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-07 07:11:47.307+0000 [id=99]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-07 07:11:47.309+0000 [id=99]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-07 07:11:47.405+0000 [id=99]    INFO    h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-08-07 07:11:47.704+0000 [id=90]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-07 07:11:47.705+0000 [id=90]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-07 07:11:47.720+0000 [id=95]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-07 07:11:47.725+0000 [id=87]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-07 07:11:47.736+0000 [id=87]    INFO    j.install.InstallState$Upgrade#applyForcedChanges: New API token system configured with insecure options to keep legacy behavior
2026-08-07 07:11:47.751+0000 [id=85]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-07 07:11:47.771+0000 [id=60]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.jna.Native in an unnamed module (file:/C:/Users/15L/.jenkins/war/WEB-INF/lib/jna-5.18.1.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled
```

4. You may also want to verify the package you downloaded. Learn more about verifying Jenkins do

Download Jenkins 2.568.2 LTS for:

Generic Java package (.war)

SHA-256: 9bbb2b329e52730ba7decd1a7a1095987f6250ec761fb21157dbb2cbcd1ef590

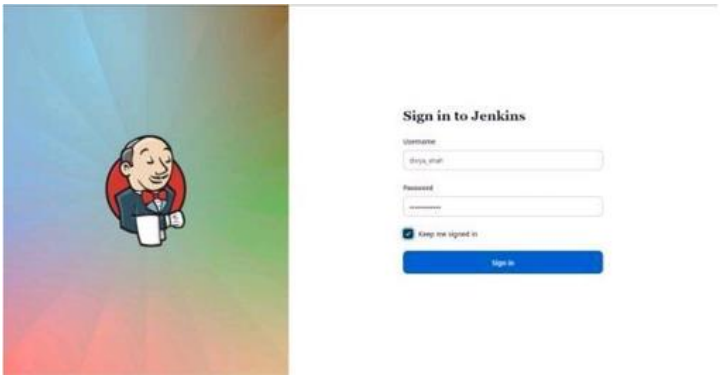
Docker

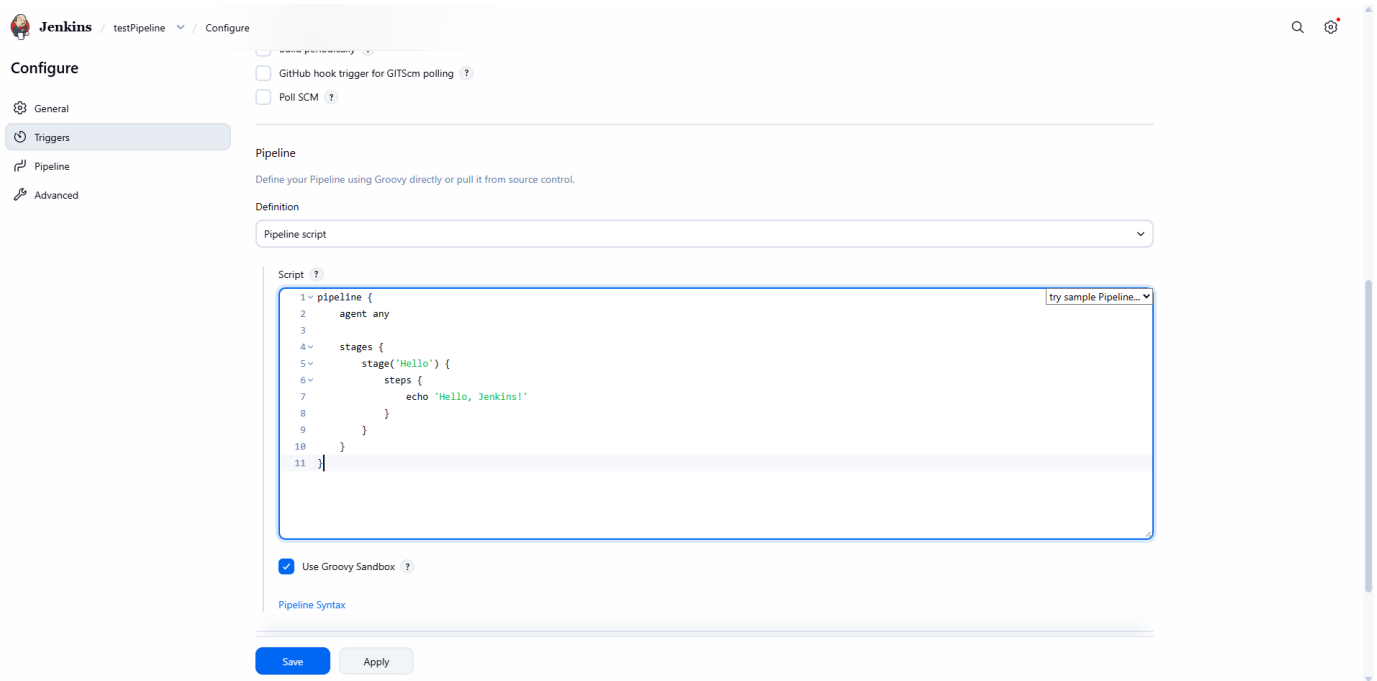
Kubernetes

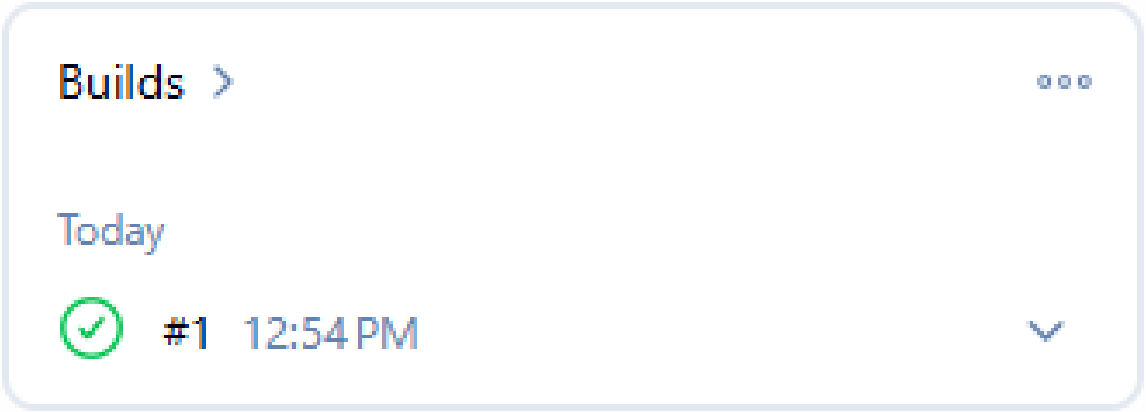
Ubuntu/Debian








Jenkins / testPipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
GitHub hook trigger for GITScm polling
Poll SCM
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
try sample Pipeline...
pipeline {
agent any
stages {
stage('Hello') {
steps {
echo 'Hello, Jenkins!'
}
}
}
}
Use Groovy Sandbox
Pipeline Syntax
Save
Apply


Builds
Today
#1
12:54 PM

```
Started by user unknown or anonymous
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\15L\.jenkins\workspace\testPipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello, Jenkins!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```groovy
node {
    stage('Hello') {
        echo 'Hello from Scripted Pipeline!'
    }
}
```

try sa

```
Started by user unknown or anonymous
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\15L\.jenkins\workspace\pipeline_57
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello from Scripted Pipeline!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Conclusion:

In this experiment, we learned the fundamentals of Continuous Integration (CI) and the role of Jenkins as an automation server in the software development lifecycle. Jenkins was successfully installed and configured, and both Declarative and Scripted Pipelines were created and executed to automate simple build tasks. The experiment demonstrated how Jenkins simplifies build automation, improves software quality through continuous integration, and supports efficient CI/CD workflows. Thus, the objectives of Learning Outcomes **LO1** and **LO3** were successfully achieved.